

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (IT) Examination****April-May 2015****IT404 Language Processor****Date: 01.05.2015, Friday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
- 5.

SECTION – I**Q - 1 Answer the following questions. [07]**

a. The output of Lexical Analyzers is _____. [01]

(a) Machine code (b) A stream of tokens

(c) Intermediate code (d) A parse tree

b. A computer program that converts whole program into machine language line by line is called _____. [01]

(a) Interpreter (b) Translator

(c) Compiler (d) All of the above

c. A loader is a program that _____. [01]

(a) Places program into main memory and prepares them for execution.

(b) Automates the translation of assembly language into machine language.

(c) Accepts program written in a high level language and produces as object program.

(d) None of these

d. In Compiler, Parsing is known as _____. [01]

(a) Lexical Analysis (b) Syntax Analysis

(c) Semantic Analysis (d) Code generation

e. Define : 1) Annotated parse tree 2) Synthesized Attribute 3) Lexeme [03]

Q – 2.a Define Language Processor. Explain language processing activities. [04]**Q – 2.b Answer any TWO questions. [10]**

(i) Explain assembly language statements in detail.

(ii) Differentiate following :

(a) Compiler and Interpreter

(b) DFA and NFA

(iii) What is macro preprocessor? Explain different types of macro with suitable example.

- Q - 3 Answer following questions. [14]**
- List out the phases of compiler. And explain role of lexical analyzer in detail. [07]
 - Explain dynamic memory allocation. [04]
 - Explain the role of semantic analysis in brief. [03]

OR

- Q - 3 Answer following questions. [14]**
- Draw NFA- $\wedge$ for following regular expression and convert it into DFA. [07]
 $(0^*/1^*)^*\#$
 - What is an activation record? Explain different fields of activation record. [04]
 - Explain single pass and multi pass compiler. [03]

SECTION – II

- Q - 4 Answer the following questions. [07]**
- The identification of common subexpression and replacement of run time computations by compile time computation is _____. [01]
 - Loop optimization
 - constant folding
 - Local optimization
 - data flow analysis
 - Shift reduce parsers are _____. [01]
 - Top down parser
 - May be top down or bottom up parser
 - Bottom up parser
 - None of the above
 - Type checking is normally done during _____. [01]
 - Lexical analysis
 - Syntax directed translation
 - Syntax analysis
 - Code optimization
 - Which of the following is not intermediate code form? [01]
 - Postfix notation
 - Three address code
 - Syntax tree
 - Quadruples
 - LR stands for _____. [01]
 - Left to right
 - Right to left scanning
 - Left to right scanning
 - Left to right scanning and right most derivation in reverse
 - _____ in which each node represents an operator and children of the node represents the operands. [01]
 - Concrete syntax tree
 - Parse Tree
 - Abstract syntax tree
 - None of the above
 - True/False: Every L-attributed grammar is S-attributed grammar. [01]

Q – 5.a Answer the following questions. [05]

(i) List and explain the conflicts in shift-reduce parsing technique. [03]

(ii) Define ambiguous grammar with example. [02]

OR

Q – 5.a List and explain the different error recovery strategies in detail. [05]

Q – 5.b Eliminate the left recursion from the following grammar: [05]

$$\begin{aligned} A &\rightarrow Ba \mid b \\ B &\rightarrow Bc \mid Ad \mid \epsilon \end{aligned}$$

OR

Q – 5.b Perform the Left factoring on following Grammar: [05]

$$\begin{aligned} E &\rightarrow T + E \mid T \\ T &\rightarrow F * T \mid F \\ F &\rightarrow id \end{aligned}$$

Q – 5.c Explain the functionality of symbol table in compiler. [04]

Q – 6.a List out different code optimization techniques and explain any one with example. [04]

Q – 6.b Find out FIRST and FOLLOW set for following grammar. [05]

$$\begin{aligned} E &\rightarrow T E' \\ E' &\rightarrow +T E' \mid \epsilon \\ T &\rightarrow F T' \\ T' &\rightarrow *F T' \mid \epsilon \\ F &\rightarrow (E) \mid id \end{aligned}$$

Q – 6.c Generate the canonical LR(0) item set for following grammar: [05]

$$S \rightarrow aSb \mid ab$$

OR

Q – 6.b What is three address code? Explain any two techniques to implement three address code. [05]

Q – 6.c Generate the canonical LR(1) item set for following grammar: [05]

$$\begin{aligned} S &\rightarrow CC \\ C &\rightarrow cC \mid d \end{aligned}$$
